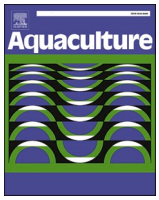
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Mathematical modelling of dissolved oxygen levels in a multi-cage salmon farm

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ABSTRACT

Dissolved oxygen concentrations were monitored at 12 points distributed over 3 depths in one out of six cages in an Atlantic salmon farm and compared to the output of a mathematical model using the advection-diffusion equation and a model of the salmon's oxygen consumption to estimate spatial oxygen distributions. The effects of oxygen consumption in neighbouring fish cages on the oxygen levels observed in the monitored cage were found to be significant, and several model variants modified to better represent the real system were tested. Position data from 8 fish in the cage equipped with acoustic telemetry tags was used to refine the model configuration.

Results show that simulating the full farm with all cages improves model fit significantly over simulating a single cage only. Even better fit was achieved in a single cage model with ad hoc modifications, indicating that there are additional non-modelled effects that should be investigated further. This study establishes the oxygen model as a suitable tool for monitoring and forecasting of oxygen conditions in salmon cages, and gives indications about he accuracy and limitations of the model.

1. Introduction

Dissolved oxygen (DO) is a key variable in Atlantic salmon sea cages, as sufficient oxygen is needed to ensure good fish welfare and high growth. The effects of a hypoxic environment range from reduced appetite, stress responses, reduced feed conversion and growth, to acute mortality, depending on how low DO levels are encountered (Oppedal et al., 2011; Remen et al., 2013, 2016; Gamperl et al., 2020). For Atlantic salmon, DO levels below 6 mg l^{-1} are generally considered hypoxic (Burt et al., 2012), although the threshold for hypoxic conditions has been found to depend on temperature (Remen et al., 2016). The risk of hypoxic conditions in a sea cage is greatest at high temperatures, when the oxygen consumption rate of the fish is high at the same time as the saturation level for DO in seawater is low. Recent studies have implied that active oxygenation of sea-cages can be a measure for reducing the risk of hypoxic conditions (Burke et al., 2022). However, to apply such methods efficiently, it is important to assess the DO level to first identify when such measures should be introduced, and then evaluate the impacts of these on the cage environment. Sensors can be used to monitor DO levels, but DO can vary widely throughout the cage volume, as well as over time, making it hard to map the spatial distribution using sensors alone (Johansson et al., 2006, 2007; Burt et al., 2012; Solstorm et al., 2018). This implies that an approach that combines real-world data from sensors with mathematical models synthesising a priori knowledge on the dynamics of the processes that determine cage DO could be called for. Developing such a concept would also represent a step towards realising a Digital Twin of the biological processes in fish farming (Føre et al., 2024).

Alver et al. (2022) presented a mathematical model of 3D distributed DO levels in salmon cages, seeking to combine our knowledge of the advective and diffusive transport of oxygen, the oxygen consumption by individual fish, and the spatial distribution of the fish. Feeding dynamics based on Alver et al. (2004, 2016) are also included in the model. The model output was validated using measured oxygen levels at Ocean Farm 1 (SalMar Aker Ocean AS, Norway), which is a very large production unit with a diameter of 110 m. Alver et al. (2023) used the model in a study investigating the effect of horizontal cage scale on DO levels, but called for further validation of the model for more common production units with diameters on the order of 50 m. The model's basic processes of advection, diffusion and oxygen consumption by fish are independent of the shape and size of the cage as long as the geometry and fish biomass is appropriately configured, but there may be

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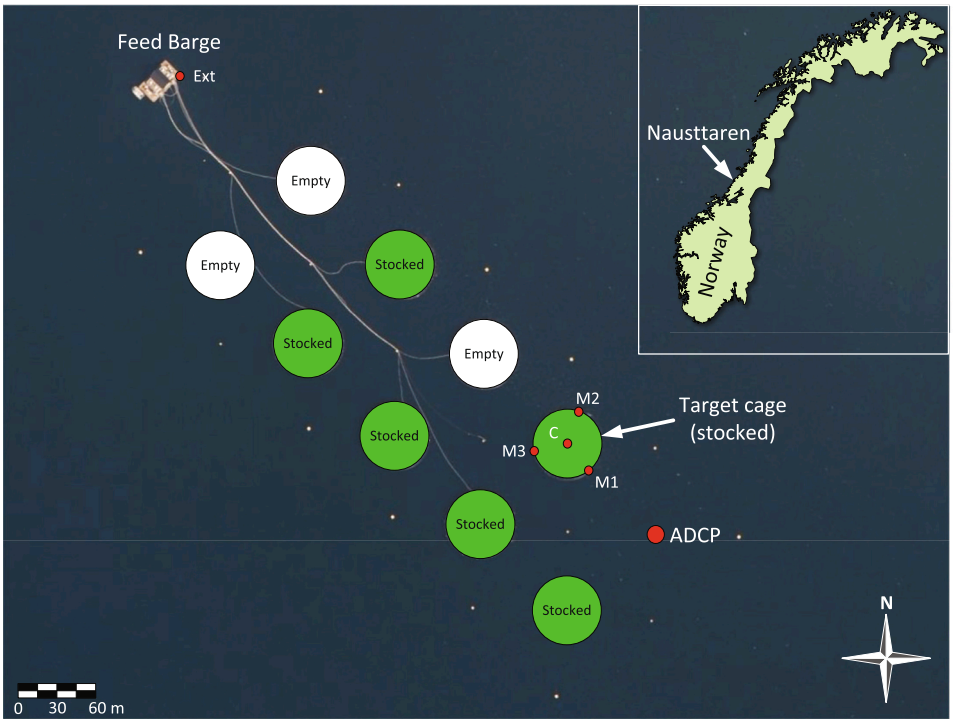


Fig. 1. Aerial view of Nausttaren with the positions of stocked/empty cages, O2 sensors (C, M1–3, Ext) and ADCP indicated (image from Norge i Bilder, © Statens kartverk, Geovekst).

systematic differences in the current conditions or fish behaviour that require model adjustments. One particular difference is that Ocean Farm 1 is operated as one large solitary unit, while smaller cages are typically placed in grid configurations with one or two rows. As a consequence, depending on the current direction, a cage may receive water that has been partly depleted of oxygen when passing through upstream cages. As the model has in previous studies been configured for single cages only, this means that the ambient oxygen levels, if measured by a sensor outside of the area, may be under- or overestimated depending on the current speed and direction.

In the original version of the model, a uniform spatial distribution was assumed for fish that were not actively feeding, while if feeding was ongoing, a given fraction of the fish was assumed to follow the spatial distribution of the feed (Alver et al., 2022). The assumption of uniform distribution is a reasonable starting point when knowledge about the actual fish distribution is sparse, but it can be a source of inaccuracy since farmed salmon have been shown to display preferences particularly in their vertical positioning (Oppedal et al., 2011). It is thus likely that estimates using the DO model could be improved if some measure of the actual fish distribution could be obtained. Several methods for observing fish in sea-cages exist, including volume targeting methods such as echo-sounders (e.g., Korsøen et al., 2009; Zhang et al., 2014; Helminen et al., 2021; Kristmundsson et al., 2023), camera-based computer vision (Saberioon et al., 2017) and biosensors/telemetry (e.g., Muñoz et al., 2020; Hvas et al., 2020; Macaulay et al., 2021). While the two former categories can provide reliable behavioural data on group and population level, biosensors/telemetry is the only method that enables the acquisition of individual histories on data. Since behaviour inherently is a process on the individual level, this method can thus shed more light on specific individual responses, but data from such tools can also be aggregated to provide insights into group dynamics. In marine sea-cages the preferred method for collecting such data is acoustic telemetry, which is seeing increased use as a research tool to study farmed fish.

In this study, the aquaculture site Nausttaren was instrumented with 15 oxygen sensors to map DO conditions in one of the six cages containing fish biomass over a nine-day period. Acoustic telemetry was used to track the 3D positions of a number of individual fish in two of the cages in the same period, thereby providing a measure of their vertical distribution, which is a key input to the mathematical model. The mathematical model is used to simulate DO levels in two ways; covering the observed cage only and covering the entire farm. By comparison to the measured values, several model modifications are tested and the fit between the model and observations is evaluated. The importance of taking the entire farm into account versus modelling a single cage is evaluated.

2. Materials and methods

2.1. Field study

2.1.1. Site description

The field study took place during 22–30 June 2022 at Nausttaren (N64.39°, E10.51°, Osen, Middle Norway), which is a full-scale Atlantic salmon farm commercially managed and operated by Bjørøya AS following typical Norwegian fish farming protocols.[1] The farm at Nausttaren is laid out in a 5 × 2 grid of 90 m × 90 m frame moorings supporting up to ten large sea-cages. At the time of this study, stocked cages were present in the 2nd and 4th position of the upper row and positions 2–5 in the lower, as shown in Fig. 1. All cages were industry-standard Polarcirkel plastic pens (AKVAgroup ASA, Norway) of cylindroconical shape with a circumference of 160 m. The depth of the upper cylindrical section was 20 m, while the bottom cone had a maximum depth of 28 m. The study's main focus was on the cage installed in the 4th position of the upper row, hereafter denoted as the target cage (see Fig. 1). The biomass of the target cage was estimated to be 487 metric tonnes (169,821 salmon of mean weight 2869.4 g) and the fish were fed during daytime 07:30–17:30 at regular time intervals, accumulating to a fixed fraction of the cage biomass per day.

[1] See Supplementary material for a photo from Nausttaren.

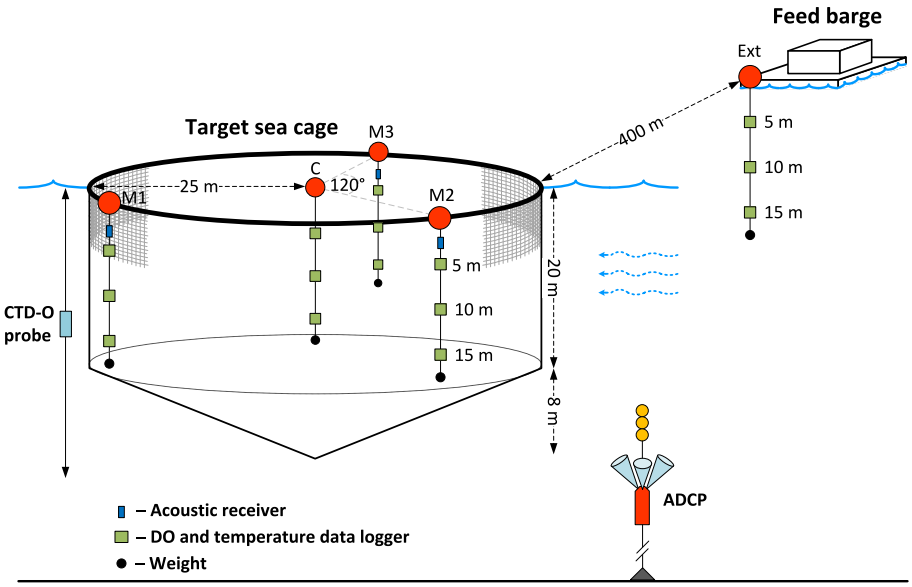


Fig. 2. Sensor placement.

2.1.2. Cage instrumentation

Time-series of dissolved oxygen and temperature were acquired from the target cage using five sensor strings, each consisting of three miniDOT data loggers (Precision Measurement Engineering, Inc., CA, USA) placed at different depths. The sensors were held in position at 5, 10 and 15 m depth by attaching them to fixed ropes that were kept vertical by heavy weights. Four sensor strings were deployed in the target cage, one in the centre of the cage (denoted C) and three evenly distributed around the perimeter of the cage (denoted M1–M3), while the fifth string was deployed from the feed barge approximately 400 m from the cages (denoted Ext) and served as an external reference measurement (Figs. 2 and 1). The time resolution of the data loggers was set to one measurement per minute. Water column profiles of salinity, temperature and dissolved oxygen were also acquired on the first and last day of the monitoring period using a CTD probe equipped with an auxiliary DO sensor (Model SD204/205, Saiv AS, Norway). Furthermore, water currents were monitored using an Aquadopp Profiler 400 kHz (Nortek AS, Norway) placed at 28 m depth looking upwards at the position indicated as ADCP in Fig. 1.

2.1.3. Acoustic telemetry

Nausttaren served as the experimental site for a separate but concurrent study on the behavioural interaction of salmon and cleaner fish, which involved acoustic tagging and tracking of the movements of a limited number of individual salmon in two of the cages. The DO sensors were installed in one of these cages (the target cage), and telemetry data was subsequently used in the model as a proxy of the spatial distribution of the fish during the monitoring period. Approximately eight months prior to the field study, a total of 16 salmon had been implanted with 9 mm acoustic tags equipped with depth sensors (D-MP9, Thelma Biotel AS, Norway), of which eight fish resided in the target cage. The fish were tagged using a standard protocol (Ulvund et al., 2021) in compliance with the Norwegian Animal Welfare Act and was approved by the Norwegian Animal Research Authority (permit ID25536, 20/188146). The individuals had an average weight of 859±201 g and length 40.1±2.7 cm at the time of tagging on 2 November 2021. No mortality was observed among the 16 fish until the cages were harvested in the end of September 2022 more than 330 days later. Each fish tag was set up to transmit depth readings 40 times per hour on average, while three time-synchronised acoustic receivers (TBR700RT, Thelma Biotel AS, Norway) were placed at fixed submerged positions evenly distributed along the perimeter of the cage to acquire the tag signals (Figs. 2). This configuration permitted 3D localisation of the tagged fish utilising the difference in signal arrival times at the receivers to calculate the horizontal coordinates and the transmitted depth as the vertical coordinate (Hassan et al., 2019).

2.1.4. Sea and weather conditions

Temperature and salinity profiles were measured down to 21 m on June 22 and July 1. On the initial day, the water column was nearly homogeneous with a temperature of 11.5 °C through the entire water column and salinity ranging from 32.3 PSU at the surface up to 32.9 PSU at 21 m. On July 1, clear stratification was seen with temperature ranging from 17 °C at the surface down to 10 °C at 21 m, and salinity ranging from 27 PSU at the surface up to 32.7 PSU at 21 m. The changes in the hydrographic profiles were likely caused by a shift from a highly wind mixed regime to a combination of weak mixing and increased heating of the surface water. The difference between low and high tide at the location was at maximum 1.82 m.

In the study period, wind speeds were strongest on June 23, around 10–15 m/s, while on all other days it mostly varied in the interval 1–10 m/s. The air temperature was mostly in the range 10–16 °C in the period except for higher temperatures in the range 14–26 °C on June 24–26.[2] The sky was cloudy on June 22 and 23, but cleared up before noon on the 24th. It gradually became more cloudy from the 25th until the afternoon of the 27th. 28th had full cloud cover, while the cloud cover was variable on the 29th and 30th.

2.2. Model structure

The model presented in Alver et al. (2022, 2023) is used in this study with a few modifications. First, input values for ambient oxygen level (ω_A), current speed (v_x and v_y) and temperature (T) are now considered depth dependent. Ambient oxygen level and temperature are based on external measurements at 5, 10 and 15 m, and vertical profiles have been estimated by linear interpolation between these three depths (values above 5 m are set equal to the values at 5 m, and values below 15 m are set equal to the values at 15 m). Current speeds are set based on the measurements from the profiling sensor.

Second, the individual oxygen consumption rate (mg/h) per kg body

[2] Wind speeds and temperatures recorded at Buholmråsa Fyr 3.5 km from Nausttaren, source https://seklima.met.no/observations/

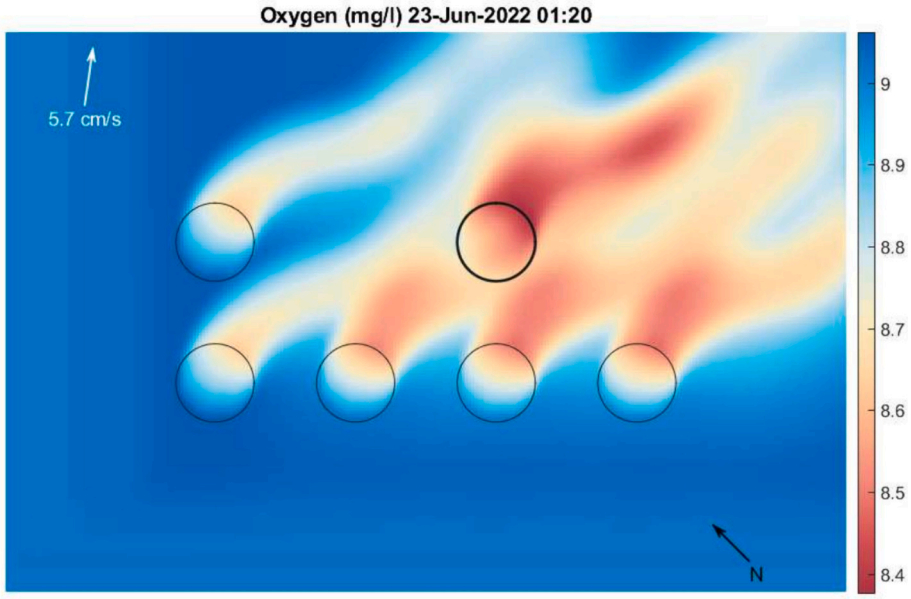


Fig. 3. Horizontal transect of model domain covering the entire farm. Cages are indicated by black circles. The observed cage is the rightmost cage in the upper row. The current speed and direction at the time of the snapshot are indicated by the white arrow. The wakes of lowered oxygen behind the cages indicate variations in the current direction in the preceding $\frac{1}{2}$ hour.

mass is still based on the empirical model derived by Grøttum and Sigholt (1998). As comparisons with measurements at Nausttaren favour a model with higher oxygen consumption, the rate has been set 20% higher than in the study at Ocean Farm 1 (Alver et al., 2022):

$$V_{O_2} = 0.0267W^{-0.33}1.03^{T}1.79^{U} \quad (1)$$

where W is body weight (kg), T is water temperature (°C) and U is swimming speed (body lengths per second).

Third, we test model modifications to account for limitations in the original model, related to 1) the vertical distribution of fish, 2) ambient oxygen levels 3) simulating the whole farm, and 4) the effect of biofouling on in-cage current speed. These are described in the following subsections.

2.2.1. *Vertical distribution of fish*

The vertical distribution of non-feeding fish can be adjusted to better correspond to the observed vertical distribution of tagged fish in the cage. This is implemented using the model's *affinity* feature, which was used to implement avoidance of low DO levels in Alver et al. (2023). Each model cell has an affinity value $a(i,j,k)$ ranging from 0 to 1 (where i, j and k are the cell coordinates), specifying the propensity of the fish to stay in or avoid the cell. 0 means complete avoidance and 1 means no avoidance. The affinity value is multiplied with the oxygen consumption value that would otherwise be used for each cell. The rates in all cells are then scaled equally to ensure the correct total oxygen consumption rate for the population.

The affinity values are set as follows:

$$a(i,j,k) = \begin{cases} 0, & \text{if outside of cage,} \\ a_v(k), & \text{if inside} \end{cases} \quad (2)$$

where $a_v(k)$ is the average relative density of the tagged fish in layer k. These values serve to restrict fish presence to the volume inside of the cage, and to skew the assumed density of fish towards the depths where the tagged fish spent the most time. The values of $a_v(k)$ are set based on the telemetry results presented in Section 3.1.

2.2.2. *Ambient oxygen levels*

The oxygen levels measured at the external sensor position may not always be representative for the oxygen level of the water flowing into the modelled cage, for two reasons. First, the water at the external sensor position may have been partly depleted of oxygen if the current flows from the farm to the sensor position.[3] Second, even though the water at the external position may be unaffected by the farm, the water flowing into the modelled cage may be partly depleted after passing through other cages.

The first source of error is mitigated by using the highest sensor value at the same depth as the ambient value, choosing between C, M1–M3 and the external position. Whenever one of the other positions records higher DO level than the external position, that level is used.

The second issue is more challenging. When the model is limited to simulating a single cage, it can be addressed in an ad hoc manner by adjusting the boundary conditions of the model given an assumption of how much the DO level will be reduced for water entering the model domain at different points. The reduction should be greatest at the edges towards the neighbouring cages and at depths with the highest density of fish. Since the DO level decreases more per m at low current speeds than at high speeds (Alver et al., 2023, Fig. 4), the reduction should be greater when current speeds are low.

The boundary values for the DO level are set based on the ambient value ω_A, and only the boundary values on the upstream edges (where water flows into the model domain) have a significant effect on the model. In the ad hoc model modification, we find the boundary value ω_b by subtracting a value from ω_A at all boundary cells at all depths that increases linearly from 0 at the east end of the domain to ω_{Red} at the west end:

$$\omega_b(i,j,k) = \omega_A(i,j,k) - \omega_{Red}(i,j,k)\cdot\frac{i_{max} - i}{i_{max}}\cdot a_v(k) \quad (3)$$

with the west-east orientation being chosen to account for the neighbouring cages being located towards the west, northwest and south as seen from the observed cage. i_{max} is the number of cells in the model domain along the west-east axis, and $a_v(k)$ is the relative density of fish in layer k used in Eq. (2). ω_{Red} is set to take higher values for low current

[3] In this study, we do not see clear indications that this occurred often in the study period (see Fig. 20 in Supplementary materials for the full time series).

Table 1
Model configuration.

Parameter	Value	Notes
Feeding times	07:30–17:30	
Feeding rate	4.83 kg/min	Total of 2900 kg over 10 h
Number of fish	169,821	Based on estimates by Bjørøya AS
Mean weight of fish	2869.4 g	Based on estimates by Bjørøya AS
Model grid resolution	2 m	
Model grid, single cage	33x33x14 cells	Cage size plus 2 × 4 cells h. and 1 cell v.
Model grid, full farm	270x180x14 cells	Farm size plus 45 cells h. and 1 cell v.
St. dev. of fish weight	573.9 kg	20% of mean weight
Time step	1 s	

Table 2
Model variants.

Model	Description
Basic	Unmodified model
V	Vertical distribution set based on telemetry data
VA	As **V**, with modification also to ambient DO levels
VAB	As **VA**, with trend in current reduction factor simulating biofouling
VBF	Full farm model, modified vertical distr., trend in current reduction

speeds:

$$\omega_{Red}(i,j,k) = \begin{cases} 0.35, & \text{if } v(i,j,k) > 6 \text{ cm/s}, \\ 1.85 - 0.25 \cdot v(i,j,k), & \text{otherwise} \end{cases} \quad (4)$$

where $v(i,j,k)$ is the current speed in cell (i,j,k).

2.2.3. Simulating the full farm

Simulating the full farm is an alternative way of obtaining more appropriate ambient DO levels for each cage. In this approach, the model is set up to cover a 4 × 2 grid of cages, disregarding the empty mooring frames at the northwest end of the farm. The model domain is oriented along the axes of the mooring grid, and with each mooring frame being 90 × 90 m, its dimensions are 540 × 360 m with the grid centered and stretching 90 m outside of the grid in all directions (Fig. 3). Since the model grid in this setup is not oriented along the north-south axis, but at an angle of 42° clockwise, measured current vectors are rotated 42° counterclockwise before being used as input to the model. For simplicity, all cages are assumed to be similar to, and contain the same biomass as, the cage of primary interest to this study. Through this setup, the model will explicitly calculate how ambient oxygen conditions are modified as water, depending on current speed and direction, passes through other cages before reaching the observed cage.

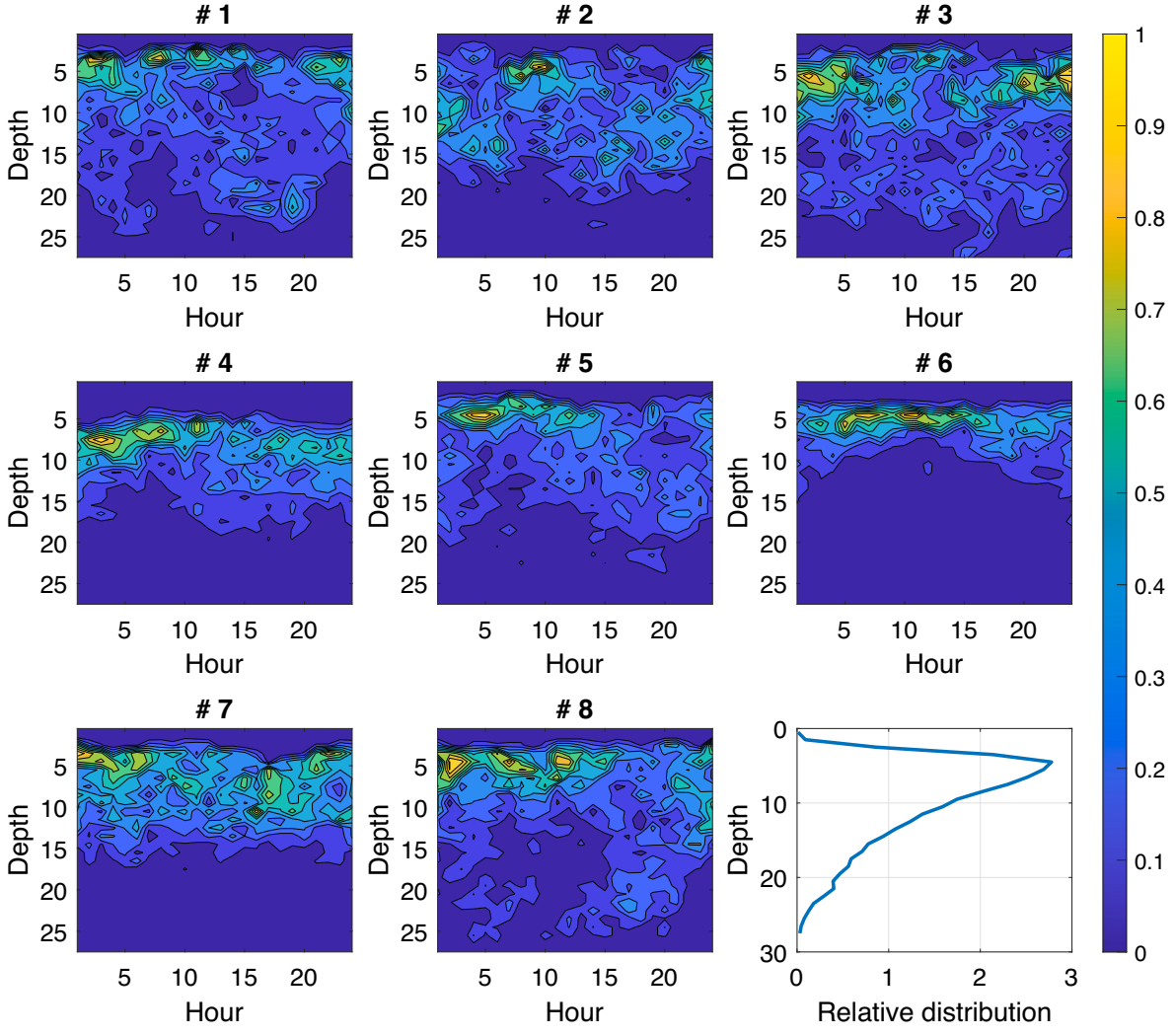


Fig. 4. Relative vertical distribution of tagged fish by time of day. The contour plots show the distribution between 1 m depth intervals of detections within each hour on any day of the study period. The lower right plot shows the relative distribution averaged over all hours and individuals.

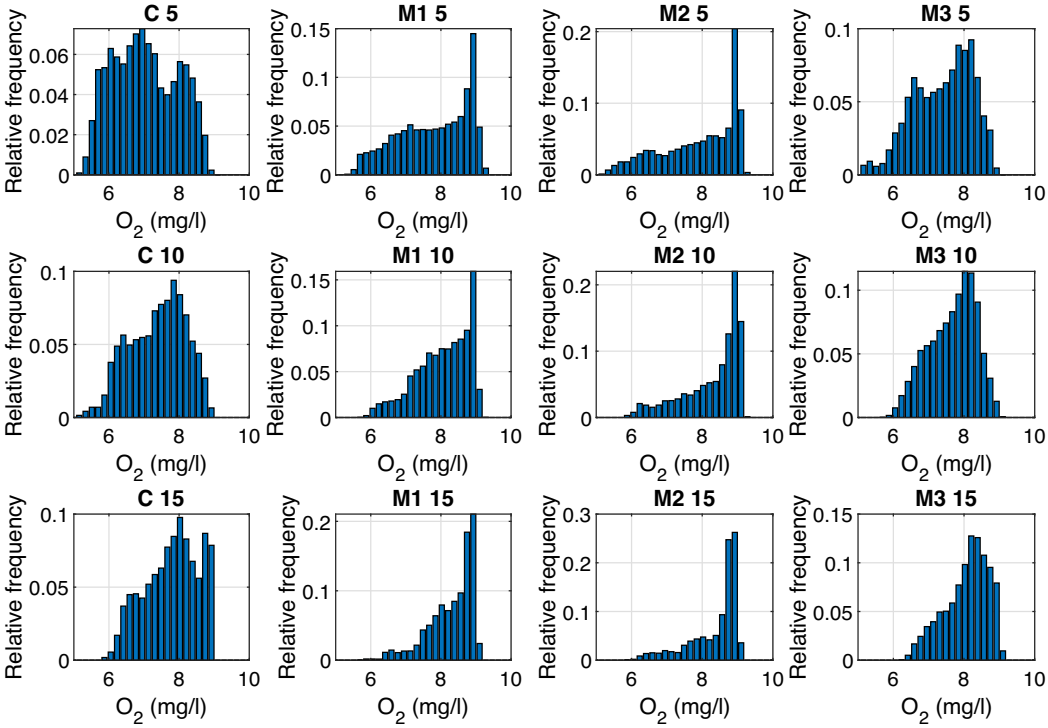


Fig. 5. Relative distributions of observed DO levels at all sensor positions over the full period of June 22 to June 30.

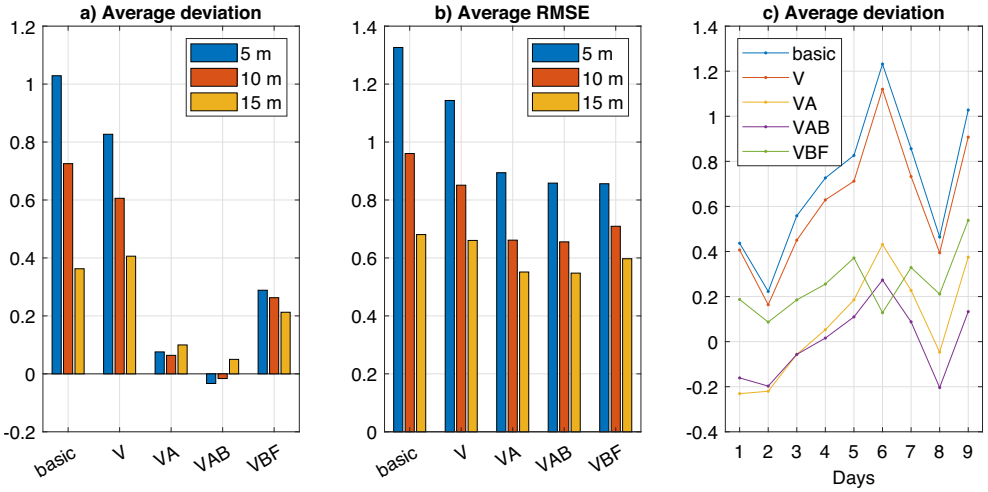


Fig. 6. a) Average deviations across sensors at each depth for the 9 day period simulated. b) Average RMSE across sensors at each depth for the same period. c) Average deviations across all sensors per day for each model variant.

2.2.4. *Effect of biofouling on current speed in cage*

In the summer season, rapid build-up of biofouling on the cage nets can be expected, even over the span of 1–2 weeks (Guenther et al., 2010), and this can significantly affect the water flow in the cages with consequences for oxygen levels (Fitridge et al., 2012; Bloecher and Floerl, 2021). Gormican (1989)measured up to 65% reduction in current speeds where cage nets were fouled by mussels and macroalgae. Cornejo et al. (2020) performed a multi-scale model study addressing the hydrodynamic effect of biofouling on salmon cages, finding reductions of up to 48% in mean flow velocities through fish cages caused by biofouling.

Biofouling is not included in our model, but its effect on flow can be simulated by varying the k_{curr} parameter that controls the reduction factor from measured ambient current speed to the speed used in the model. We use a default value of $k_{curr} = 0.8$ as in Alver et al. (2022), but to simulate a gradual build-up of moderate biofouling we test a gradual reduction of k_{curr} from 0.85 on the first day to 0.65 on the 9th day.

2.3. *Model configuration*

The model configuration is summarized in Table 1. The modelled cage has a radius of 25 m and a depth of 25 m, and the model domain is set up to include the cage plus 4 cells beyond its radius horizontally, and the cage depth plus 1 cell vertically. Each cell is classified as inside the cage if its center is 25 m from the center of the domain or closer, and 25 m from the surface or closer.

Five variants of the model are tested over the study period of 9 days, applying different selections of the modifications described above, and one of them with a model domain covering the full farm (Table 2). In addition, a number of tests were run to investigate potential sources of

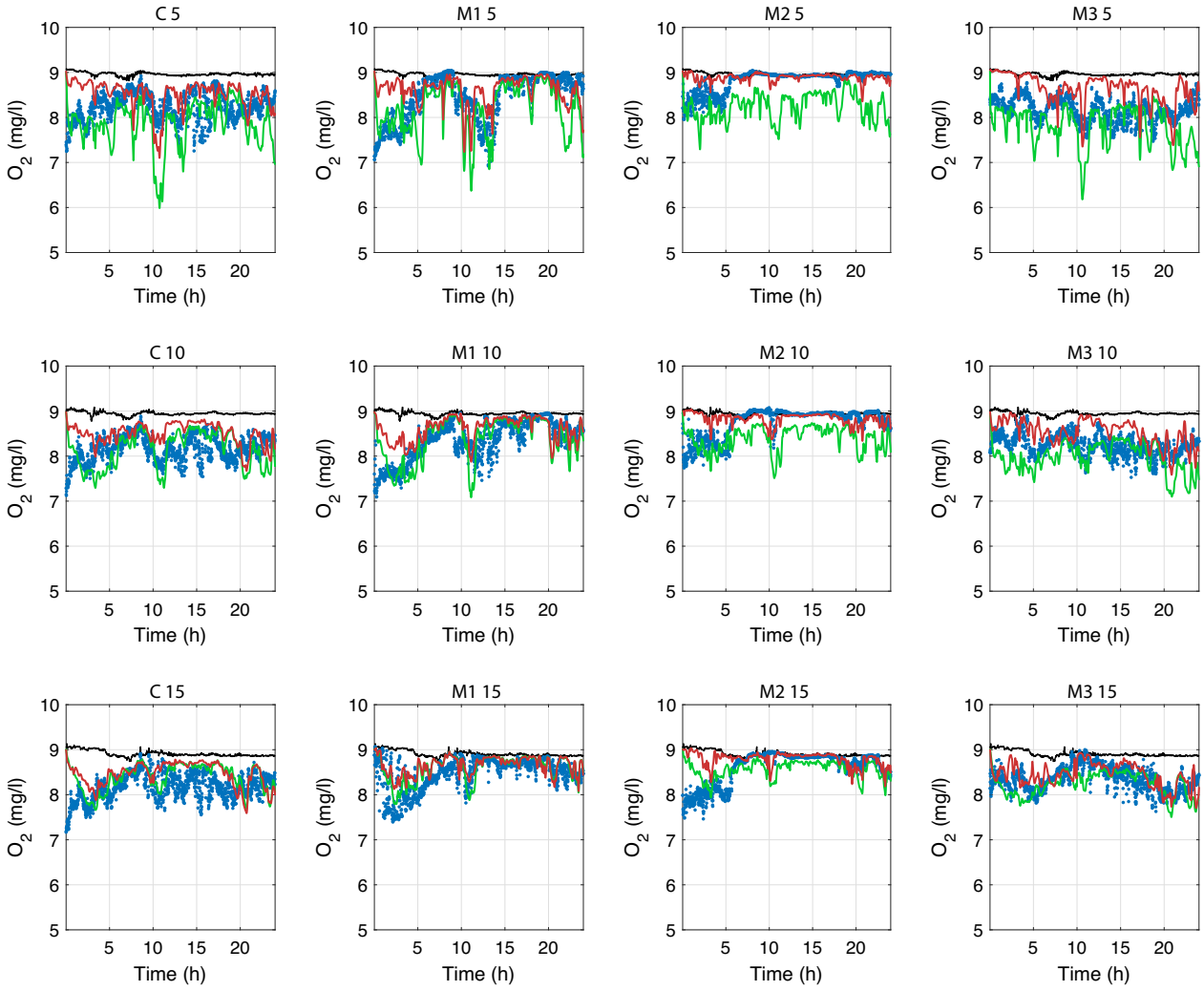


Fig. 7. Observed DO levels (blue dots) compared to **VAB** model (green) and **basic** model (red) at the 12 sensor positions on the 23rd of June. Assumed ambient DO levels at the three depths are shown as black lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

model error, such as uncertainty in current speed and direction, horizontal distribution of fish, diffusivity parameters, overall oxygen consumption rate and temperatures, with the objective to improve our understanding of model errors and their potential causes. The tests are summarized along with their results in Section 3.4.

2.4. Model evaluation

Agreement between observed and modelled DO levels at the 12 sensor positions (C and M1–M3) is used as a basis for comparing model variants. For each sensor position, the model value is recorded as the concentration in the cell containing the position at the times corresponding to each measurement time. Visual comparison over 9 days for 12 sensors is difficult, so model fit has been quantified using average deviation and Root Mean Square Error (RMSE), computed for all sensor positions and averaged across positions. For the average deviation, positive or negative values close to 0 are desired, and for the non-negative RMSE lower values are better.

3. Results

3.1. Spatial distribution of tagged fish

The acoustic receiver that obtained the most registrations throughout the experimental period recorded a total of 33,107 data points for the eight tagged fish.

The recorded depth values of the tagged fish in the cage show clear individual differences, but also some common patterns (Fig. 4). Overall, the fish spend most time around 5 m depth, with a sharp decline from this depth towards the surface. Below 5 m, the time spent at each depth decreases gradually, with few registrations below 25 m depth. Averaging the distribution over all hours and individuals (Fig. 4, lower right panel) gives a relatively smooth distribution with a peak around 5 m. This distribution is used to set the values of $a_v(k)$ in Eq. (2).

Horizontal positions were also recorded, with results indicating that the fish utilize the horizontal space relatively uniformly, with a slight bias towards the central parts of the cage (see plots in Supplementary material).

3.2. Observed range of DO levels

Observed DO levels at all positions vary within the interval 5–10 mg l^{-1} over the study period (Fig. 5 and Supplementary material), The center positions experience levels in the lower part of the interval more frequently than the edge positions. The M1 and M2 sensors experience levels near ambient DO levels relatively frequently compared to the M3 sensors, which are located closest to the center of the farm and see somewhat reduced levels most of the time. DO levels are generally

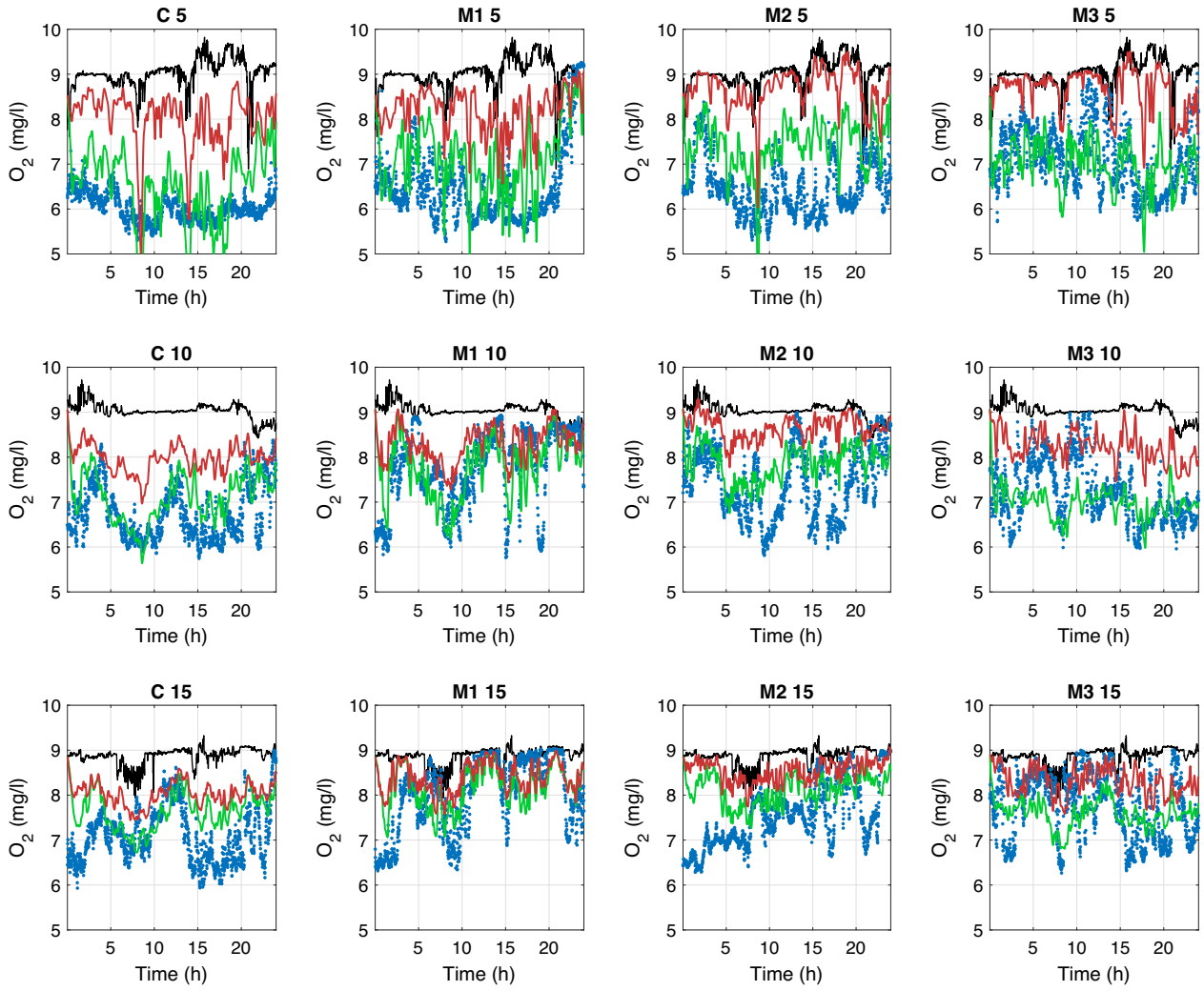


Fig. 8. Observed DO levels (blue dots) compared to **VAB** model (green) and **basic** model (red) at the 12 sensor positions on the 27th of June. Assumed ambient DO levels at the three depths are shown as black lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

lowest at 5 m and highest at 15 m, which is consistent with the indications of the depth distribution of fish from the telemetry data. At 10 and 15 m, levels are rarely below 6 mg l^{-1}, while this occurs more often at 5 m, particularly at the center position.

3.3. Comparison between modelled and observed values

The overall model fit is better on some days than others, and varies between sensor positions. To compare the model variants we therefore look at the metrics of model deviation, representing the bias in modelled values compared to observed values, and RMSE, representing the typical scale of deviations in the absolute sense. The model variants are compared in terms of these metrics, averaged for all sensors at each observation depth over the period from June 22nd to 30th, in Fig. 6a and b. Average deviation across all sensors is shown per day to indicate trends over time in Fig. 6c. The **basic** model has a clear positive bias, particularly at 5 m, and there is a positive trend in the bias over the 9 day period. The **V** model reduces the bias only slightly, **VBF** reduces it by two thirds, **VA** reduces it further and **VAB** has approximately 0 bias. The average RMSE is reduced slightly by **V** compared to **basic**, and further reduced by other variants. The trend in bias over time is clearly positive for all model variants except **VAB**, for which it is approximately neutral.

For the **basic** model, the most positive bias is seen at 5 m, and the smallest, but still positive, bias is seen at 15 m. The **V** model reduces bias at 5 m and 10 m, and increases it at 15 m, so the differences by depth are reduced. The **VA** model both reduces the overall bias significantly, and counters the differences by depth. The **VAB** model has near 0 overall bias. The full farm simulation **VBF** shows a nearly similar positive bias at all depths. For all model variants, RMSE values are highest at 5 m and lowest at 15 m, indicating that the overall model fit is better at higher depth for all model variants.

To give an impression of the variability between sensor positions and over time, Fig. 7 shows time series for the **VAB** and **basic** models from June 23, where the model fit was best, and Fig. 8 shows June 27, where the poorest fit was seen. Fig. 9 shows a comparison of the **VAB** and **VBF** models on June 27. See Supplementary material for comparisons covering all days of the study, and for rose plots showing distributions of observed current speeds and directions on all days at three depths. Current speeds were lower on June 27 than on June 23, leading to generally lower DO levels, as seen at the center position in the cage. The directional distribution is relatively similar at 11 m, but at 5 and 15 m the fraction of directions recorded within the north-east quadrant is clearly higher on June 27, which means more downstream effects from nearby cages are to be expected compared to June 23.

On June 23, measured DO levels are generally high, and most of the time the reduction from ambient level is less than 1.5 mg l^{-1}. Both models generally reproduce the measured levels. There are some events where the models predict strong reductions in DO level that are not seen

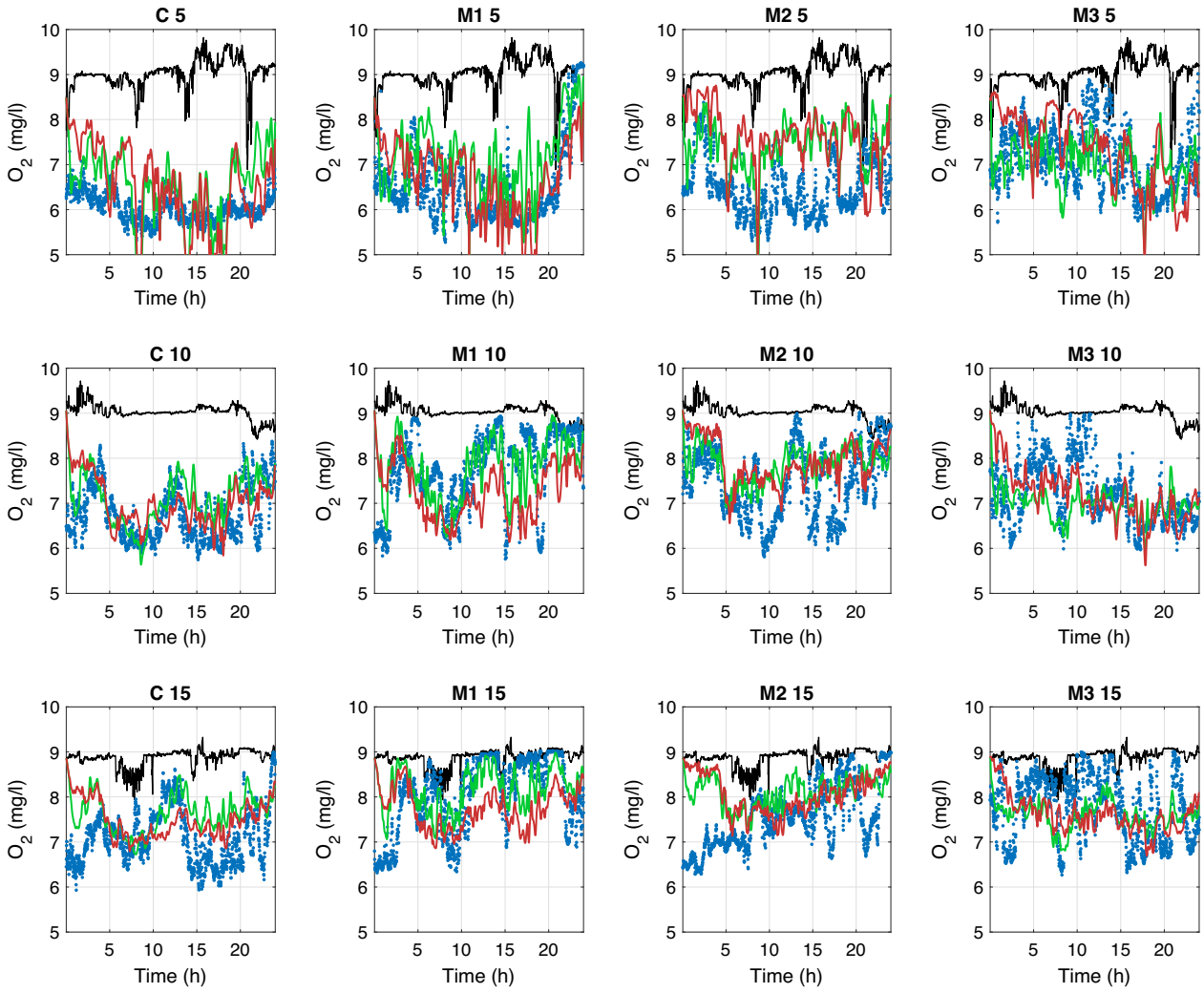


Fig. 9. Observed DO levels (blue dots) compared to **VAB** model (green) and full farm **VBF** model (red) at the 12 sensor positions on the 27th of June. Assumed ambient DO levels at the three depths are shown as black lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in observations. In the period 10–17 h, the **VA** model underestimates the level at the M2 locations due to the ad hoc reduction in ambient values, while the **VBF** model correctly predicts levels close to the ambient level. In the same time period, **VA** predictions are close to observed values at the M3 locations, while the **VBF** model overestimates levels.

On June 27, DO levels are lower, with reductions from ambient levels ranging from 0 to 4 mg l^{-1}. Both models predict overall similar values to the observations at some sensor locations. **VA** overestimates DO at C sensors, M1 5 m, M2 5 and 15 m, while **VBF** overestimates at M1 5 m and underestimates at M1 10 and 15 m.

3.4. *Sensitivity to uncertain inputs and parameters*

Tables 3 and 4 summarize tests made to investigate potential sources of error in the model with the aim to uncover the system's sensitivity to the factors tested. DO levels at 5 m on June 27 for the three tests with the clearest effects are shown in Fig. 10, compared with the levels of the **VA** model. The current direction test (# 1) gives markedly reduced deviations at the M2 position, but the fit becomes much worse particularly at M1. The horizontal fish distribution test (# 2) gives smaller changes, and the largest deviations remain large. The current speed test (# 3) results in significantly lower DO levels at some times, and leads to the model clearly underestimating levels e.g. at C and M1, while the positive bias e.g. at M2 during hours 12–20 is not notably reduced.

4. Discussion

The results from the five model variants show that the oxygen model is useful in describing DO dynamics in a typical open Atlantic salmon sea cage, provided adaptations are made to account for the effect of neighbouring cages and the vertical distribution of the fish. The oxygen model is a simplified representation of a highly variable system affected by several interacting factors, so perfect agreement with the real system is not to be expected. Furthermore, the field observations are also associated with uncertainty and may not fully correspond to the real system. In particular, ACDP measurements have higher relative uncertainty at low current speeds, and events with low current speeds also pose the highest risk of seeing low DO levels in the cage.

The field observations present a consistent picture where DO levels at the measurement positions along the edge of the cage and in the center range from near ambient levels to significantly lower levels, with generally lower values near the surface compared to deeper parts of the cage, and generally lower levels in the center compared to the edges. The variation with depth is consistent with telemetry data indicating that the fish spend the most time around 5 m depth, implying that the highest oxygen consumption rate will occur around this depth. The lower levels at the center positions are expected, since the water reaching these sensors will normally have passed through part of the cage to be partly depleted of oxygen. Since the observed cage has

Table 3
A selection of error sources in the model, and how these have been tested and evaluated (Part 1).

#	Potential source of error	Test description	Outcome
1	Uncertainty in current directions around and inside cage	**VA** model for June 27 with current direction overridden so M2 sensors are always directly downstream.	Significant changes particularly at M1 and M2 positions. Average reduction of 0.93, 0.80 and 0.45 mg l^{-1} at M2 5, 10 and 15 m, respectively. Periodically wrong current direction could potentially explain large deviations. However, while deviations become reduced at M2 5 m, they become much larger at M1 5 m, so there is no consistent improvement.
2	Uncertainty in horizontal distribution of fish	**VA** model for June 27 with cage divided horizontally in two halves, with one half centered on the M3 sensors. All non-feeding fish are confined to the opposite half to simulate a highly skewed fish distribution.	DO levels decreased on average by 0.27 mg l^{-1} at the M1 positions and 0.19 mg l^{-1} at M2, and increased by 0.33 mg l^{-1} at M3. The differences are not large enough to counter deviations seen at the M1 and M2 positions.
3	The internal current speed may be lower than 0.8 times the external current speed	**VA** model for June 27 with current reduction factor lowered from 0.8 to 0.5.	Significant reduction averaging 0.35–1.55 mg l^{-1} at sensor positions, leading to significant underestimation of DO for long periods at many sensor positions. The clearest positive deviations, e.g. at M2 5 m (Fig. 8), are still not fully countered. The overall effect is significantly worse model fit.
4	Boundary effect, water leaving and reentering domain is reset to ambient DO level.	**V** model for all days with increased domain size (2.5 radius lengths beyond cage wall in all directions).	Slight improvement in model fit, with 10% reduced bias and 5% reduced RMSE. This indicates that domain boundaries should be placed some distance from the edges of the cage.

Table 4
A selection of error sources in the model, and how these have been tested and evaluated (Part 2).

#	Potential source of error	Test description	Outcome
5	Uncertainty in horizontal and vertical diffusivity coefficients.	**VA** for all days with 5 times higher/lower vertical diffusivity coefficient.	Primary effect is slightly higher/lower DO levels at 5 m, so average bias is increased/decreased. Average RMSE is unchanged/increased.
		VA with 5 times higher/lower horizontal diffusivity	Very small effects.
6	Uncertainty in oxygen consumption rate of fish	**VA** model for all days with global 20% increase in oxygen consumption.	On average, approximately 0.1 mg l^{-1} reduction at sensor positions. No significant effect on large deviations, and worse overall fit in terms of RMSE.
7	Temperatures above 5 m may be underestimated	**VA** model for all days with 3°C added to temperature values at 5 m before vertical interpolation.	Very small effect, on average <0.1 mg l^{-1} reduction at 5 m sensors, and less at 10 and 15 m.
8	Model resolution	**VA** model for all days with 1 m resolution.	Minor differences in DO levels at sensor positions, and no improvement in model fit compared to 2 m resolution.

neighbouring cages where oxygen is being consumed at similar rates, it is also as expected that the sensors in the M3 positions, closest to the center of the farm, almost all the time record reduced DO levels compared to ambient levels.

The positioning of the external sensors in such a study is a challenge, both for the ADCP and the external oxygen sensors, when the purpose is to measure ambient conditions. The locations of the external oxygen sensors (by the farm's feed barge) and the ADCP (at an empty mooring frame in the corner of the farm) were chosen partly out of convenience, as they needed anchoring points to maintain fixed positions. As the fish cages affect both oxygen levels and the flow field, positions further away from the fish biomass would give readings less affected by the farm, but on the other hand, relatively close positions are needed to ensure that the measurements are representative for the fish cages we are observing. For operational use of the oxygen model, the external sensor positions used in this study are representative of what one can expect to have access to in a commercial farm.

The telemetry data was also used as input to the model, and while the devices themselves are known to provide reliable depth measurements to within their specified accuracy, only a few individuals of the population was equipped with tags. This could in principle lead to inaccurate estimates of the swimming depth distribution if by chance the selected animals had behaviours that deviated from that seen in the main bulk of the population. While this is factor to always be aware of, all tagged individuals in this study exhibited similar behavioural patterns. This implies that the tagged fish were a part of the 'main body' of the population, suggesting that it is unlikely that the low number and selection of tagged fish had a large impact on model performance. Another concern that can be raised when using such tools can be that the handling/surgery involved in deploying acoustic telemetry tags can have adverse impacts on fih state, and thereby their behaviour (Macaulay et al., 2021). However, the procedure applied in this case is a well-known and widely used method that has been proven to have little impact on the fish post-tagging through several previous studies (e.g., Urke et al., 2013; Bjerck et al., 2021; Ulvund et al., 2021; Hassan et al., 2022), and is thus unlikely to have had a large impact on the observed behaviour. This assumption is further strengthened by that the fish had been tagged almost eight months prior to the study period and all tagged fish not only survived but showed active and seemingly normal movements within the cage until the fish was slaughtered three months later.

Two factors are particularly challenging to model: the spatial distribution of fish within the cage can deviate significantly from the model's assumptions over shorter or longer time intervals, and the effect of nearby cages on the DO level of water flowing into the cage is hard to quantify. Three model modifications have been tested, addressing these factors, and all of them lead to better model fit in terms of average RMSE.

Minor improvement was gained by adjusting the vertical distribution of the fish based on acoustic telemetry data, suggesting that the original assumptions made for the distribution of the fish were suboptimal. This modification reduced the disparity where model bias was highest at 5 m and lowest at 15 m.

The greatest RMSE improvement was gained from adjusting the ambient DO level along the west-east axis, and this adjustment also significantly reduced the average positive bias seen in the model compared to observations. This suggests that the effect of nearby cages is

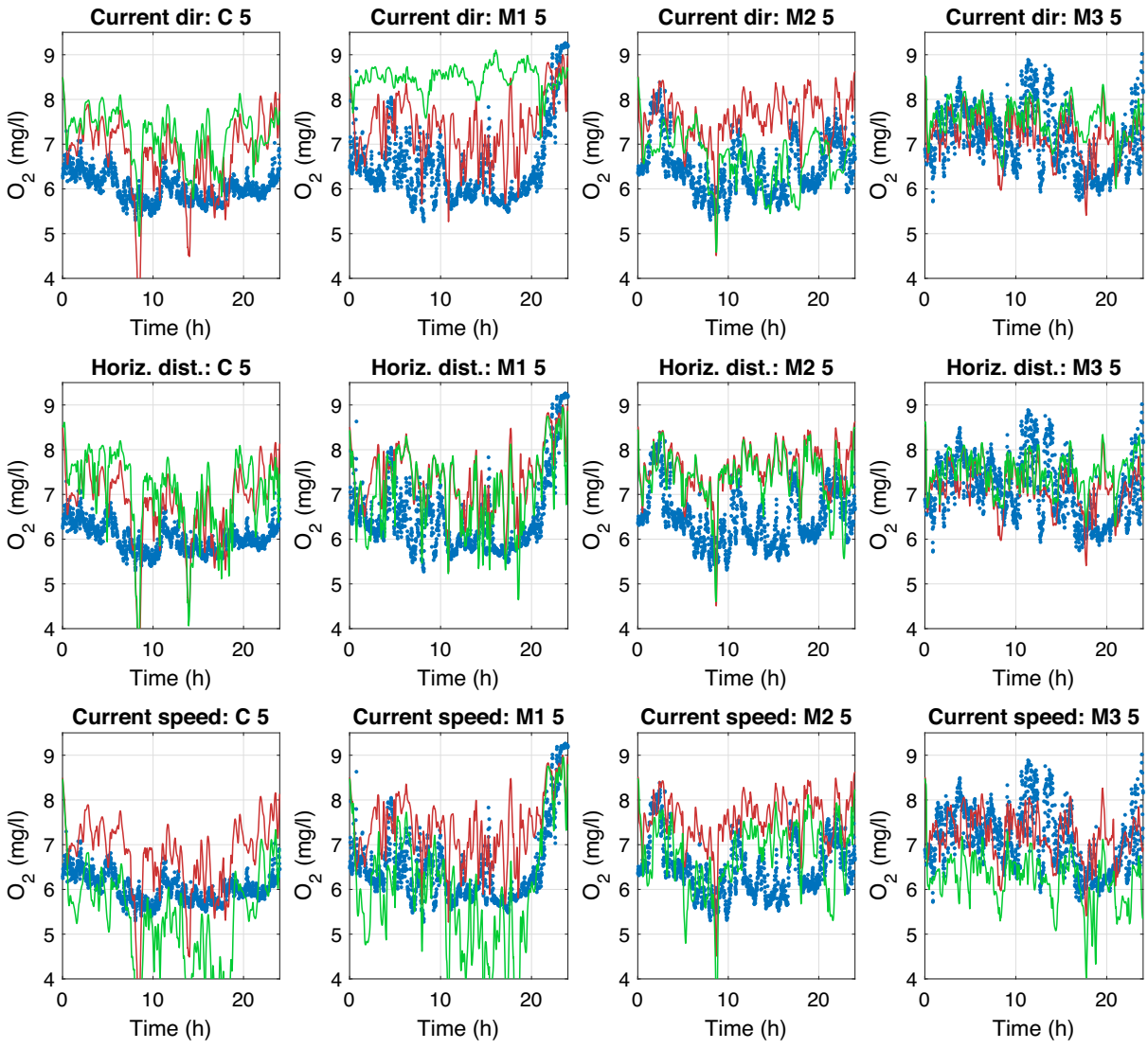


Fig. 10. Observed DO levels (blue dots) compared to test case simulations from Table 3 (green) and **VA** model (red) at 5 m sensor positions. Sub titles indicate the tests and sensor positions shown. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

a significant source of error in the **basic** model. As the adjustment made in the **VA** model was fairly simplistic, we could reasonably expect significant further gains in accuracy if the ambient levels around the cage were better known. The drawback of this approach is its ad hoc nature, requiring detailed knowledge of the particular conditions surrounding the modelled cage to configure the model. Furthermore, there is no simple way to calculate how much the boundary values should be reduced at each point. The ambient value adjustment along with the change in vertical distribution removed the disparity in model bias with depth, as well as reducing the overall bias.

Simulating the full farm is a more robust approach to handling the effect of neighbouring cages, since it only requires knowledge of the physical layout of the farm. The **VBF** model performs clearly better than the **basic** and **V** models, although the model fit is not as good as for the **VA** and **VAB** models. Despite bias and RMSE improvements and an overall good match with observations, there are clear deviations between the modelled and observed DO levels. That the **VA** model performs better than **VBF** indicates that the ad hoc modification of ambient DO levels compensates not only for the effect of neighbouring cages, but also for other factors that lead to the ambient DO level being overestimated. One such factor may be the boundary effect tested in test 4 of Table 3. If current directions are fluctuating, water may leave and re-enter the model domain, and thereby become reset to ambient DO level. By moving the model boundary further out from the cage, this effect can be reduced, and the test did indicate a slight improvement when testing this for the **V** model.

It is also worth commenting on the computational cost of expanding the model to cover the full farm. In the present scenario, this increases the number of model cells by a factor of 45 for an equal resolution, leading to a similar increase in computational load. This makes the model less convenient to run, but model parallelization can significantly reduce the simulation time of the expanded model when sufficient CPU cores are available.

When testing a time dependent current reduction factor representing gradually increased biofouling on the net cages in the **VAB** and **VBF** models, the overall bias was reduced, and the positive trend in the bias over the experimental period was countered. The trend is likely to be caused by some unmodelled factor that changed throughout the study period, and the results from these models indicate that biofouling may have been that factor. This could harmonise with the findings of Bi and Xu (2018) and others who have explored the impact of biofouled nets on the attenuation of the current after passing through the net wall. Specifically, Bi and Xu (2018) found that heavily fouled nets could induce an attenuation as high as 51 to 71%, which is higher than the

attenuation rates assumed in the present study.

The details of the flow field within a cage or farm have not been investigated in this study. Flow fields within and around fish cages are hard to assess due to the difficulty of measuring inside cages filled with fish. It is clear that the cages and the biomass within them have an effect on the local flow field (Michelsen et al., 2019; Jónsdóttir et al., 2021), but no quantitative model of the 3D flow field within a cage has been established. CFD simulation, e.g. as made by Broch et al. (2020), can provide estimates that have a higher degree of realism than the assumption of uniform flow field that has been applied in this study. We have run model tests using such a flow field as part of this study, but the effect on model performance was relatively minor, and will therefore not be described in further detail here. Still, more detailed flow fields is an interesting avenue for further research on the oxygen dynamics of salmon cages.

5. Conclusion

In this study, we have investigated the dynamic DO conditions in an Atlantic salmon aquaculture farm with open sea cages though field studies and several variants of a mathematical model. The results show that the model predicts the overall spatial and temporal variations in DO levels well, although with deviations occurring due to a number of known and unknown unmodelled effects. Important conditions that can be taken into account in the model are the effect of each fish cage on the DO levels in other cages of the farm, the vertical distribution of the fish and the the flow rate within the farm, possibly affected by the biofouling state of the nets.

Further studies are needed to address deviations that cannot be explained by known factors, and the internal flow field within cages might be one important target for further research. For high accuracy operational estimation of DO levels in salmon cages, a combination of measurements and modelling is likely to be needed. Achievement of this goal would also largely provide a foundation for realising one of the industrially relevant case studies for Digital Twins in aquaculture presented by Føre et al. (2024), which could lead to the model having even larger future impacts.

CRediT authorship contribution statement

Morten Omholt Alver: Conceptualization, Methodology, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Martin Føre:** Methodology, Writing – review & editing. **Henning André Urke:** Methodology, Investigation, Writing – review & editing. **Jo Arve Alfredsen:** Methodology, Investigation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Jo Arve Alfredsen reports equipment, drugs, or supplies was provided by Bjørøya Oppdrett AS. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material to this article can be found online at https://doi.org/10.1016/j.aquaculture.2024.741291.

References

Alver, M.O., Alfredsen, J.A., Sigholt, T., 2004. Dynamic modelling of pellet distribution in Atlantic salmon (*Salmo salar* L.) cages. Aquac. Eng. 31, 51–72. https://doi.org/10.1016/j.aquaeng.2004.01.002.

Alver, M.O., Skøien, K.R., Føre, M., Aas, T.S., Oehme, M., Alfredsen, J.A., 2016. Modelling of surface and 3d pellet distribution in Atlantic salmon (*Salmo salar L.*) cages. Aquac. Eng. 72, 20–29. https://doi.org/10.1016/j.aquaeng.2016.03.003.

Alver, M.O., Føre, M., Alfredsen, J.A., 2022. Predicting oxygen levels in Atlantic salmon (*Salmo salar*) sea cages. Aquaculture 548, 737720. https://doi.org/10.1016/j.aquaculture.2021.737720.

Alver, M.O., Føre, M., Alfredsen, J.A., 2023. Effect of cage size on oxygen levels in Atlantic salmon sea cages: a model study. Aquaculture 562, 738831. https://doi.org/10.1016/j.aquaculture.2022.738831.

Bi, C.W., Xu, T.J., 2018. Numerical study on the flow field around a fish farm in tidal current. Turk. J. Fish. Aquat. Sci. 18, 705–716. https://doi.org/10.4194/1303-2712-v18_5_06.

Bjerck, H.B., Urke, H.A., Haugen, T.O., Alfredsen, J.A., Ulvund, J.B., Kristensen, T., 2021. Synchrony and multimodality in the timing of Atlantic salmon smolt migration in two norwegian fjords. Sci. Rep. 11, 6504. https://doi.org/10.1038/s41598-021-85941-9.

Bloecher, N., Floerl, O., 2021. Towards cost-effective biofouling management in salmon aquaculture: a strategic outlook. Rev. Aquac. 13, 783–795. https://doi.org/10.1111/raq.12498.

Broch, O.J., Klebert, P., Michelsen, F.A., Alver, M.O., 2020. Multiscale modelling of cage effects on the transport of effluents from open aquaculture systems. PLoS One 15, e0228502. https://doi.org/10.1371/journal.pone.0228502.

Burke, M., Grant, J., Filgueira, R., Swanson, A., 2022. Oxygenation effects on temperature and dissolved oxygen at a commercial Atlantic salmon farm. Aquac. Eng. 99, 102287. https://doi.org/10.1016/j.aquaeng.2022.102287.

Burt, K., Hamoutene, D., Mabrouk, G., Lang, C., Puestow, T., Drover, D., Losier, R., Page, F., 2012. Environmental conditions and occurrence of hypoxia within production cages of Atlantic salmon on the south coast of newfoundland. Aquac. Res. 43, 607–620. https://doi.org/10.1111/j.1365-2109.2011.02867.x.

Cornejo, P., Guerrero, N.M., Montes, R.M., Quiñones, R.A., Sepúlveda, H.H., 2020. Hydrodynamic effect of biofouling in fish cage aquaculture netting. Aquaculture 526, 735367. https://doi.org/10.1016/j.aquaculture.2020.735367.

Fitridge, I., Dempster, T., Guenther, J., De Nys, R., 2012. The impact and control of biofouling in marine aquaculture: a review. Biofouling 28, 649–669. https://doi.org/10.1080/08927014.2012.700478.

Føre, M., Alver, M.O., Alfredsen, J.A., Rasheed, A., Hukkelås, T., Bjelland, H.V., Su, B., Ohrem, S.J., Kelasidi, E., Norton, T., et al., 2024. Digital twins in intensive aquaculture—challenges, opportunities and future prospects. Comput. Electron. Agric. 218, 108676. https://doi.org/10.1016/j.compag.2024.108676.

Gamperl, A.K., Ajiboye, O.O., Zanuzzo, F.S., Sandrelli, R.M., de Fátima, C., Peroni, E., Beemelmanns, A., 2020. The impacts of increasing temperature and moderate hypoxia on the production characteristics, cardiac morphology and haematology of Atlantic salmon (*Salmo salar*). Aquaculture 519, 734874. https://doi.org/10.1016/j.aquaculture.2019.734874.

Gormican, S.J., 1989. Water circulation, dissolved oxygen, and ammonia concentrations in fish-net cages. Ph.D. thesis. University of British Columbia doi: 10.14288/1.0097472. URL: https://open.library.ubc.ca/collections/ubctheses/831/items/1.0097472.

Grøttum, J.A., Sigholt, T., 1998. A model for oxygen consumption of Atlantic salmon (Salmo salar) based on measurements of individual fish in a tunnel respirometer. Aquac. Eng. 17, 241–251. https://doi.org/10.1016/S0144-8609(98)00012-0.

Guenther, J., Misimi, E., Sunde, L.M., 2010. The development of biofouling, particularly the hydroid ectopleura larynx, on commercial salmon cage nets in mid-Norway. Aquaculture 300, 120–127. https://doi.org/10.1016/j.aquaculture.2010.01.005.

Hassan, W., Føre, M., Urke, H.A., Kristensen, T., Ulvund, J.B., Alfredsen, J.A., 2019. System for Real-Time Positioning and Monitoring of Fish in Commercial Marine Farms Based on Acoustic Telemetry and Internet of Fish (IoF), in: Proceedings of the 29th International Ocean and Polar Engineering Conference, Honolulu, Hawaii, USA (pp. ISOPE–I).

Hassan, W., Føre, M., Urke, H.A., Ulvund, J.B., Bendiksen, E., Alfredsen, J.A., 2022. New concept for measuring swimming speed of free-ranging fish using acoustic telemetry and doppler analysis. Biosyst. Eng. 220, 103–113. https://doi.org/10.1016/j.biosystemseng.2022.05.013.

Helminen, J., O'Sullivan, A.M., Linnansaari, T., 2021. Measuring tailbeat frequencies of three fish species from adaptive resolution imaging sonar data. Trans. Am. Fish. Soc. 150, 627–636. https://doi.org/10.1002/tafs.10318.

Hvas, M., Folkedal, O., Oppedal, F., 2020. Heart rate bio-loggers as welfare indicators in Atlantic salmon (*Salmo salar*) aquaculture. Aquaculture 529, 735630. https://doi.org/10.1016/j.aquaculture.2020.735630.

Johansson, D., Ruohonen, K., Kiessling, A., Oppedal, F., Stiansen, J.E., Kelly, M., Juell, J. E., 2006. Effect of environmental factors on swimming depth preferences of Atlantic salmon (*Salmo salar L.*) and temporal and spatial variations in oxygen levels in sea cages at a fjord site. Aquaculture 254, 594–605. https://doi.org/10.1016/j.aquaculture.2005.10.029.

Johansson, D., Juell, J.E., Oppedal, F., Stiansen, J.E., Ruohonen, K., 2007. The influence of the pycnocline and cage resistance on current flow, oxygen flux and swimming behaviour of Atlantic salmon (*Salmo salar L.*) in production cages. Aquaculture 265, 271–287. https://doi.org/10.1016/j.aquaculture.2006.12.047.

Jónsdóttir, K.E., Volent, Z., Alfredsen, J.A., 2021. Current flow and dissolved oxygen in a full-scale stocked fish-cage with and without lice shielding skirts. Appl. Ocean Res. 108, 102509. https://doi.org/10.1016/j.apor.2020.102509.

Korsøen, Ø.J., Dempster, T., Fjelldal, P.G., Oppedal, F., Kristiansen, T.S., 2009. Long-term culture of Atlantic salmon (*Salmo salar L.*) in submerged cages during winter affects behaviour, growth and condition. Aquaculture 296, 373–381. https://doi.org/10.1016/j.aquaculture.2009.08.036.

Kristmundsson, J., Patursson, Ø., Potter, J., Xin, Q., 2023. Fish monitoring in aquaculture using multibeam echosounders and machine learning. IEEE Access. https://doi.org/10.1109/ACCESS.2023.3320949.

Macaulay, G., Warren-Myers, F., Barrett, L.T., Oppedal, F., Føre, M., Dempster, T., 2021. Tag use to monitor fish behaviour in aquaculture: a review of benefits, problems and solutions. Rev. Aquac. 13, 1565–1582. https://doi.org/10.1111/raq.12534.

Michelsen, F.A., Klebert, P., Broch, O.J., Alver, M.O., 2019. Impacts of fish farm structures with biomass on water currents: a case study from frøya. J. Sea Res. 154, 101806. https://doi.org/10.1016/j.seares.2019.101806.

Muñoz, L., Aspillaga, E., Palmer, M., Saraiva, J.L., Arechavala-Lopez, P., 2020. Acoustic telemetry: a tool to monitor fish swimming behavior in sea-cage aquaculture. Front. Mar. Sci. 7, 645. https://doi.org/10.3389/fmars.2020.00645.

Oppedal, F., Dempster, T., Stien, L.H., 2011. Environmental drivers of Atlantic salmon behaviour in sea-cages: a review. Aquaculture 311, 1–18. https://doi.org/10.1016/j.aquaculture.2010.11.020.

Remen, M., Oppedal, F., Imsland, A.K., Olsen, R.E., Torgersen, T., 2013. Hypoxia tolerance thresholds for post-smolt Atlantic salmon: dependency of temperature and hypoxia acclimation. Aquaculture 416-417, 41–47. https://doi.org/10.1016/j.aquaculture.2013.08.024.

Remen, M., Sievers, M., Torgersen, T., Oppedal, F., 2016. The oxygen threshold for maximal feed intake of Atlantic salmon post-smolts is highly temperature-dependent. Aquaculture 464, 582–592. https://doi.org/10.1016/j.aquaculture.2016.07.037.

Saberioon, M., Gholizadeh, A., Cisar, P., Pautsina, A., Urban, J., 2017. Application of machine vision systems in aquaculture with emphasis on fish: state-of-the-art and key issues. Rev. Aquac. 9, 369–387. https://doi.org/10.1111/raq.12143.

Solstorm, D., Oldham, T., Solstorm, F., Klebert, P., Stien, L.H., Vågseth, T., Oppedal, F., 2018. Dissolved oxygen variability in a commercial sea-cage exposes farmed Atlantic salmon to growth limiting conditions. Aquaculture 486, 122–129. https://doi.org/10.1016/j.aquaculture.2017.12.008.

Ulvund, J., Engebretsen, S., Alfredsen, J., Kristensen, T., Urke, H., Jansen, P., 2021. Behavioural response of farmed Atlantic salmon (*Salmo salar L.*) to artificial underwater lights: wavelet analysis of acoustic telemetry data. Aquac. Eng. 95, 102196. https://doi.org/10.1016/j.aquaeng.2021.102196.

Urke, H., Kristensen, T., Ulvund, J., Alfredsen, J., 2013. Riverine and fjord migration of wild and hatchery-reared Atlantic salmon smolts. Fish. Manag. Ecol. 20, 544–552. https://doi.org/10.1111/fme.12042.

Zhang, H., Wei, Q., Kang, M., 2014. Measurement of swimming pattern and body length of cultured chinese sturgeon by use of imaging sonar. Aquaculture 434, 184–187. https://doi.org/10.1016/j.aquaculture.2014.08.024.